# AN ELEMENT OF THE PRODUCT OF THE LOCALIZATIONS AT THE MINIMAL PRIMES THAT DIFFERS FROM AN ELEMENT OF THE MAXIMAL IDEAL BY A NILPOTENT-TYPE TERM HAS A POWER IN THE IMAGE OF THE RING

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted span cites two cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

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*Allowed standard prerequisites (editorial).*

- Existence of an element generating the radical of the maximal ideal in a one-dimensional local ring. Stacks Project, Algebra, Lemma algebra-lemma-height-1 (tag 00OK). Provides g with m_A = radical of (g), which realizes L as the localization A_g.
- Zero-dimensional rings as products of localizations at minimal primes. Stacks Project, Algebra, Proposition algebra-proposition-dimension-zero-ring (tag 00JA). Justifies the identification L = A_g used to clear denominators.

*Source reference.* Original reference 00KK: sources/stacks-chow/tags.txt, source lines 754–754. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 00KJ: sources/stacks-chow/tags.txt, source lines 753–753. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $A$ be a Noetherian local ring of dimension 1. Let $L = \prod A_{\mathfrak{p}}$ and $I = \bigcap \mathfrak{p}$ where the product and intersection are over the minimal primes of $A$. Let $f \in L$ be an element of the form $f = i + a$ where $a \in \mathfrak{m}_A$ and $i \in IL$. Then some power of $f$ is in the image of $A \to L$.*

*Proof.* Since $A$ is Noetherian we have $I^t = 0$ for some $t > 0$. Suppose that we know that $f = a + i$ with $i \in I^k L$. Then $f^n = a^n + na^{n-1}i \bmod I^{k+1}L$. Hence it suffices to show that $na^{n-1}i$ is in the image of $I^k \to I^k L$ for some $n \gg 0$. To see this, pick a $g \in A$ such that $\mathfrak{m}_A = \sqrt{(g)}$ (Algebra, Lemma 00KK). Then $L = A_g$ for example by Algebra, Proposition 00KJ. On the other hand, there is an $n$ such that $a^n \in (g)$. Hence we can clear denominators for elements of $L$ by multiplying by a high power of $a$. □

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